

MATH 212

HOMEWORK DUE WEEK 9 FRIDAY

Winter 2026

Prof. Chowdhury

Note: Each question is worth 6 points.

Some of the questions requires you to write scripts. Please create a single notebook file that includes the code for all the question.

I might add one more problem at the end, but if I do, that will act as a replacement for the last lab.

■ Question 9001.

□

In this exercise, we are going to examine a very common scientific situation: we have data from an experiment and a (challenging to solve analytically) differential equation modeling the data that has some unknown parameters. We want to find the value of the parameter that gives us the best fit between our numerical solution and the experimentally observed data.

As an example, let $T(t)$ denote the temperature of an object in a vacuum chamber at time t . By the Stefan-Boltzmann Law of Radiative Cooling, the rate of change of the temperature is proportional to the difference between the fourth powers of the object's temperature and the constant ambient temperature T_{env} . This is modeled by the initial value problem:

$$\frac{dT}{dt} = -k[T(t)^4 - T_{env}^4], \quad T(t_0) = T_0$$

where $k > 0$ is an unknown constant specific to the object's radiative properties.

Assume an ambient temperature of $T_{env} = 3.0$. You are provided with a comma-separated values (CSV) file, `radiative_cooling_data.csv` in Canvas. This file contains a dataset $\mathcal{D} = \{(t_i, T_i)\}_{i=0}^N$ of experimental observations of the object's temperature over time.

- (a) Import the experimental data from `radiative_cooling_data.csv` into your computational environment.*
- (b) Use the RK4 routine you created in the last exercise to approximate the solution to the IVP (that uses only the experimental observation at t_0).*

Your implementation must accept the parameter k as its sole input and output a discrete sequence of approximated temperatures $y_i(k) \approx T(t_i)$ corresponding to the time steps t_i in the dataset.

- (c) Experiment with different k values and plot the approximations against the experimental dataset $\mathcal{D}$ until they somewhat resemble each other. You probably want to start around 10^{-5} .*

Based on these results, establish an interval $[k_{\min}, k_{\max}]$ that most likely contains the true parameter k .

- (d) Formulate the objective function $S : \mathbb{R}^+ \rightarrow \mathbb{R}$ defined by the sum of squared residuals between the numerical approximation and the experimental data:*

$$S(k) = \sum_{i=0}^N (y_i(k) - T_i)^2$$

Implement a script to evaluate $S(k)$ for a list of k values in $[k_{\min}, k_{\max}]$ with appropriately chosen Δk . Evaluate and plot $S(k)$ over your chosen interval to verify convexity in this region.

(e) Utilize an established numerical optimization routine from a standard library to minimize $S(k)$.

Note: In Python, you can use `scipy.optimize.minimize` and in Mathematica, you can use `FindMinimum`.

(f) Report the optimal parameter k^* . Superimpose your optimized numerical solution $y_i(k^*)$ onto the scatter plot of the experimental dataset to visually verify the fit.

At this point you can now use the best numerical solution to answer questions about the scientific setup (e.g. extrapolation).

(g) Approximately how long will it take for the temperature to be reduced to 5?

■ Question 9002.

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Consider the IVP

$$y'(t) = 2ty(t), \quad y(0) = 1, \quad 0 \leq t \leq 1,$$

with exact solution $y(t) = e^{t^2}$.

Let $h = 1/N$ with $N \in \mathbb{N}$, and define $t_n = nh$. We write $y_n \approx y(t_n)$.

(a) Define $f(t, y) = 2ty$. The two-step Adams–Bashforth method is

$$y_{n+1} = y_n + h \left(\frac{3}{2} f(t_n, y_n) - \frac{1}{2} f(t_{n-1}, y_{n-1}) \right), \quad n \geq 1.$$

Rewrite and simplify the recurrence in the context of our ODE.

(b) This method needs two starting values. Use*

$$y_0 = 1, \quad y_1 = e^{h^2}.$$

Implement the method and compute the approximation

$$Y_h := y_N \approx y(1).$$

For $h = 2^{-k}$ with $k = 3, 4, \dots, 10$, compute the error

$$E_h := |Y_h - e|.$$

Make a log–log plot of $\log(E_h)$ versus $\log(h)$.

*Here we are intentionally removing start-up error so we can focus on the multistep method.

- (c) Use the plot to estimate the slope p where the data look close to a line. Report your estimate of p . Or use some kind of in built linear regression command.
- (d) Assume that for sufficiently small h the numerical result has an asymptotic expansion of the form

$$Y_h = y(1) + Ch^2 + Dh^3 + \mathcal{O}(h^4),$$

where C, D do not depend on h .

Let $Y_{h/2}$ be the value produced by the same method on the same problem using step size $h/2$ (so it takes $2N$ steps to reach $t = 1$). We would like to form a weighted average

$$\widetilde{Y}_h := a Y_{h/2} + b Y_h$$

so that $\widetilde{Y}_h$ still converges to $y(1)$ as $h \rightarrow 0$, but has a smaller leading error term.

- (i) Explain why it is necessary that $a + b = 1$.
- (ii) Choose a and b so that $\widetilde{Y}_h = y(1) + \mathcal{O}(h^3)$. You should get specific rational numbers.
- (e) Compute

$$\widetilde{E}_h := |\widetilde{Y}_h - e|$$

for the same step sizes $h = 2^{-k}$, $k = 3, \dots, 10$. Make a second log-log plot of $\log(\widetilde{E}_h)$ versus $\log(h)$ and estimate the new slope. Compare with your estimate from part (d).

- (f) Using your data, give a numerical estimate for the constant C in

$$Y_h - y(1) \approx Ch^2$$

by fitting a line to $\log(E_h)$ vs. $\log(h)$ and using the intercept. Then explain, using the expansion in part (e), why your choice of a, b removes the Ch^2 term.

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The method described above is called Richardson's Extrapolation. It can be used for most Numerical Methods to increase the order of the error. Look it up online for more examples.

■ Question 9003.

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Consider the linear system $A\vec{x} = \vec{b}$ with

$$A = \begin{pmatrix} 10 & -9 & 0 & 0 \\ 0 & 10 & -9 & 0 \\ 0 & 0 & 10 & -9 \\ -1 & -1 & -1 & 10 \end{pmatrix}, \quad \vec{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \quad \vec{x}^{(0)} = \vec{0}.$$

Define the permutation that reorders the unknowns as

$$\vec{\tilde{x}} = \begin{pmatrix} x_4 \\ x_3 \\ x_2 \\ x_1 \end{pmatrix} = P\vec{x}, \quad P = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Form the permuted system

$$\tilde{A} \tilde{\mathbf{x}} = \tilde{\mathbf{b}}, \quad \tilde{A} = PAP^T, \quad \tilde{\mathbf{b}} = P\mathbf{b}.$$

- (a) By modifying the Jacobi code from class or your own implementation, run Jacobi on $A\mathbf{x} = \mathbf{b}$ for 25 iterations and record (and plot) the residual norms

$$\|\vec{r}^{(k)}\|_\infty, \quad \vec{r}^{(k)} = \mathbf{b} - A\mathbf{x}^{(k)}.$$

- (b) Run Jacobi on the permuted system $\tilde{A} \tilde{\mathbf{x}} = \tilde{\mathbf{b}}$ with $\tilde{\mathbf{x}}^{(0)} = \vec{0}$ for 25 iterations and record (and plot)

$$\|\tilde{\vec{r}}^{(k)}\|_\infty, \quad \tilde{\vec{r}}^{(k)} = \tilde{\mathbf{b}} - \tilde{A} \tilde{\mathbf{x}}^{(k)}.$$

Compare your plot to part (a). What do you observe?

- (c) Prove that the error for the Jacobi method remains invariant under the permutation of the columns of A . In other words,

$$\|\tilde{\vec{r}}^{(k)}\|_\infty = \|\vec{r}^{(k)}\|_\infty \quad \text{for all } k.$$

- (d) By modifying the Gauss-Seidel code from class or your own implementation, run Gauss-Seidel on $A\mathbf{x} = \mathbf{b}$ for 25 iterations (with $\mathbf{x}^{(0)} = \vec{0}$) and plot $\|\vec{r}^{(k)}\|_\infty$.

- (e) Run Gauss-Seidel on the permuted system $\tilde{A} \tilde{\mathbf{x}} = \tilde{\mathbf{b}}$ for 25 iterations (with $\tilde{\mathbf{x}}^{(0)} = \vec{0}$) and plot $\|\tilde{\vec{r}}^{(k)}\|_\infty$. Compare your Gauss-Seidel plots from parts (d) and (e). Which ordering converges faster?

- (f) Write Gauss-Seidel in the form

$$\mathbf{x}^{(k+1)} = T_{GS} \mathbf{x}^{(k)} + \vec{c}_{GS}, \quad T_{GS} = -(D + L)^{-1}U.$$

Compute T_{GS} for A and $\tilde{T}_{GS}$ for $\tilde{A}$ (you should use Mathematica or some similar programming language), then compute $\rho(T_{GS})$ and $\rho(\tilde{T}_{GS})$ numerically. Explain how these values account for what you observed in parts (d) and (e).

- (g) In a short paragraph, explain why the conclusion of part (c) does not generally hold for Gauss-Seidel.

■ Question 9004.

□

Note: I am making this an optional exercise. You do not need to submit it, but you should try to prove it yourself.

Prove that the operator norms induced by the ℓ_∞ -norm satisfies

$$\|A\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}|. \quad (\text{maximum row sum})$$